

ECE 647: Lecture 2

Not all formulated problems are straightforward to come up with a solution, and some are more difficult than others.

We specifically focus on the **convex problems**.

1 Convex problems

Some features of *convex problems* are as follows:

- Control variables are real numbers (not integers)
- Objective function and constraints are continuous. E.g., not step function
- Objective function and constraints are convex

Such problems are called *convex optimization problems*. We are interested in convex problems because of the following reasons:

- They can model many (not all) interesting problems.
- There are well-studied mechanisms to develop efficient and reliable solutions.

Useful technical features

- Local minimum is also the global minimum
- No duality gap. The optimal solution can also be arrived at by solving the dual problem.
- Dual problem may lead to distributed solutions

Many of these conveniences are due to convexity.

However, it is not straightforward to formulate a problem in a convex form.

- More of an art than science
- There are tons of examples of success

Goal of this class

The goal of this class in the *first part* is as follows:

- Understand the characteristics of convex problems and common approaches to solving them
- Gain some research in translating research problems into convex programs and solve them.
- Understand the issues and challenges to develop practical protocols based on convex optimization.

The class will cover the following.

- **Fundamentals of theory**
 - Convexity
 - Duality
 - Convergence
- **Applications in Communications and Networking**
 - **PHY/MAC**: Power control, scheduling, and random access.
 - **Transport**: Congestion control, TCP
 - **Routing**: Multi-path routing, network flow
 - cross-layer design for wireless communication networks
- **Applications in Statistical Learning**
 - Regression
 - Classification

We will use these applications to illustrate how convex optimization is used in practical problems.

The *second part* of this class will be on *stochastic optimization* under uncertainty.

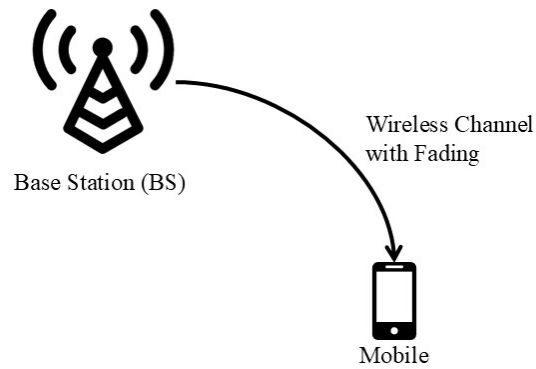
1.1 Stochastic Optimization/MDP

We will study the **MDP - Markov Decision Program**

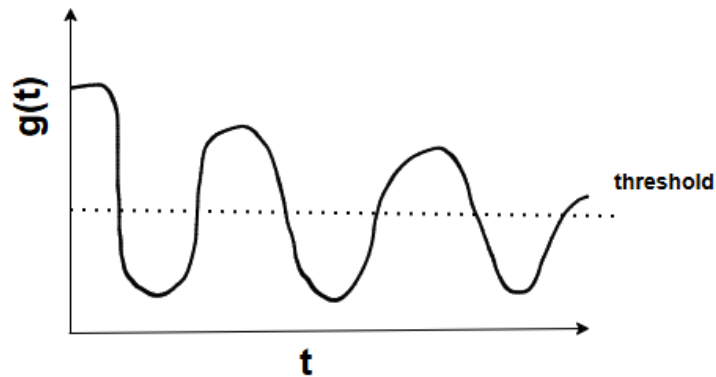
- Decisions are typically taken in stages
- Significant uncertainty in each stage
 - . In the **Payoff** or **reward** - E.g., transmission rate depends on random channel conditions
 - . In the **State** of *next stage* - E.g., Random channel errors impact whether transmission succeeds or not
- Decisions in one stage directly affect decisions in the next stages.
 - If a packet is transmitted successfully in the current stage, the *backoff* in the future stages will be lower
- MDP gives a way to solve for per-stage state-dependent decisions for maximizing the **reward**.

Although convex optimization problems can handle uncertainty, we need MDP when the objectives and constraints are more stringent.

- E.g., a base station wishes to transmit the maximum amount of data subject to some average power constraint

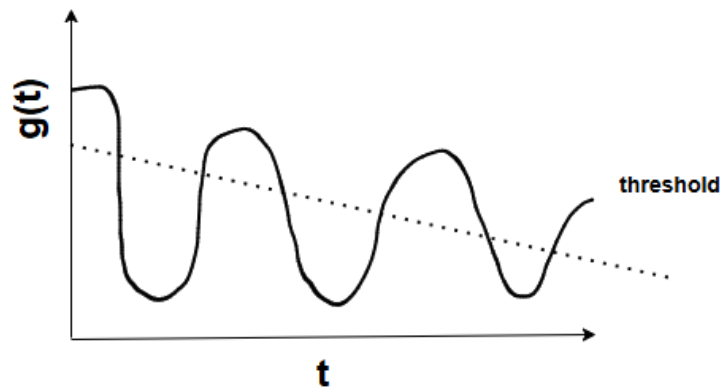


- when we consider an infinite time horizon, this can be formulated as a convex optimization problem, as shown earlier.
- However, if we impose the more stringent constraint that the energy must be spent within τ time-slots, then MDP will be needed.
- **Why?**
 - The overall idea of any optimal decision is to spend energy when the channel is *good*.
 - One can imagine that there is a threshold for *good* channel conditions



- with infinite time-horizon, every time-slot statistically faces the same situation
 - infinite number of future slots

- infinite amount of total energy to spend
- Thus, we expect that this threshold is independent of time-slot and independent of how much energy has been spent in the past.
- when there are only τ slots, however,
 - The closer we are to the end, the more *urgent* we are to spending the remaining energy.
 - The more energy remaining, the more *urgent* to spend it.
 - Otherwise, we may have to *waste* the energy in the last time-slot, whose channel could be really bad.
- As a result, the threshold will likely depend on both time and the state (i.e., the energy budget left).



- In summary, we expect the first case to be *easier*.
 - Convex optimization problems with a few variables while the second case is *harder*
 - MDP with state-/stage-dependent decisions
- We will study the foundation theory of MDP and how to use it to solve research problems.
- The first set of lectures will introduce the following:
 - Convex sets
 - Convex functions

- Convex optimization problems minimize a convex function over a convex set
- Familiarity with these concepts is important because it helps us quickly recognize if we are on the path to formulating a convex optimization problem.
- Let us first build the knowledge towards **convex sets**.

1.2 Sets

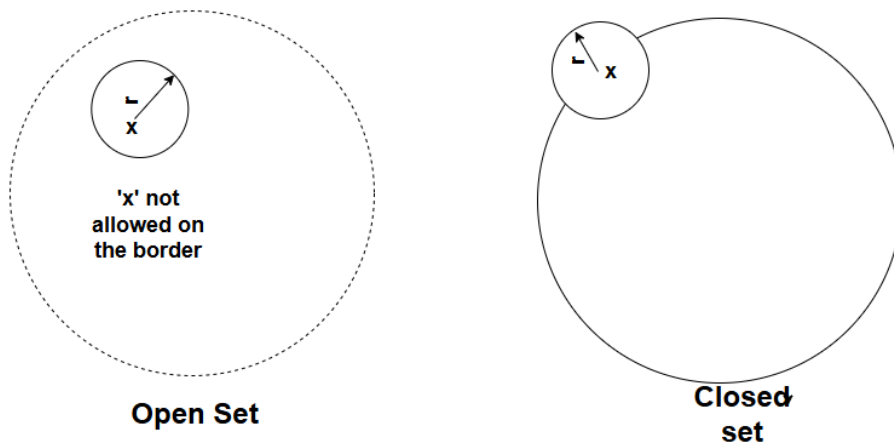
- We often work in the space $\mathbb{R}^n$, $x \in \mathbb{R}^n$ where $x = (x_0, x_1, \dots, x_n)^T$
- **Euclidean norm** - $\|x\| = \sqrt{x^T x} = \sqrt{\sum_{i=1}^n x_i^2}$

Open Set $\mathcal{O}$

- No point in $\mathcal{O}$ is on the *border*
- More precisely, define a ball centered at x

$$B(x, r) = \{y : \|y - x\| \leq r\} \quad (1)$$

Definition: A set $\mathcal{O}$ is *open* if for any $x \in \mathcal{O}$, there exists $r > 0$ such that $B(x, r) \subseteq \mathcal{O}$



Examples of open sets:

- $\|x\| < a$
- $\|x\| > a$
- Open interval $(0,1)$ is an open set

Closed Set $\mathcal{F}$

A set $\mathcal{F}$ is a closed set if $\bar{\mathcal{F}}$ is an open set. Since an open set does not contain a boundary, its complement will.

Examples of closed sets:

- $\|x\| \leq r$
 - $\|x\| \geq r$
 - Closed interval $[0, 1]$ is a closed set
- Any intersection of closed sets is still a closed set
 - Any union of open sets is still an open set

Closure - $\text{cl } \mathcal{A}$ - Smallest closed set containing $\mathcal{A}$

Interior - $\text{int } \mathcal{A}$ - The largest open set contained in $\mathcal{A}$

Boundary - The boundary of a set $\mathcal{A}$ is given as $\text{cl } \mathcal{A} \setminus \text{int } \mathcal{A}$

- Many important results of convex theory apply to both open and closed sets
- In some cases, convexity can replace the need to assume whether a set is closed or open (see later).
- Even though we are more interested in convex sets, there is some close relation between convex and affine sets:
 - e.g., separation theorems
 - , which is very important for convex theory.

- In addition, affine sets are convex sets.

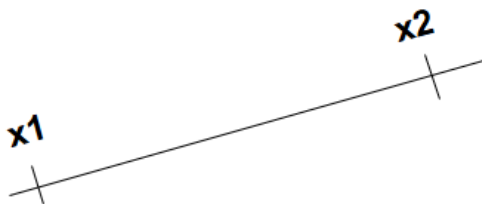
Affine sets

A set $\mathcal{C}$ is an affine set if for any $x_1, x_2, \dots, x_k \in \mathcal{C}$, then the linear combination $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in \mathcal{C} \forall \theta_1 + \theta_2 + \dots + \theta_k = 1$

Examples of *affine sets*:

1. A **1-D affine set** - Suppose $x_1, x_2 \in \mathcal{C}$. For $\mathcal{C}$ to be an affine set, we must have

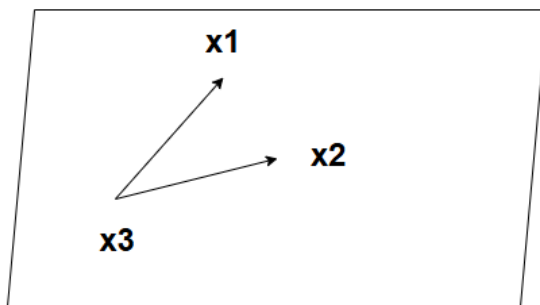
$$\begin{aligned} \theta_1 x_1 + \theta_2 x_2 &\in \mathcal{C}, \forall (\theta_1 + \theta_2) = 1 \\ \implies \theta_1 x_1 + (1 - \theta_1) x_2 &\in \mathcal{C}, \\ \implies x_2 + \theta_1 (x_1 - x_2) &\in \mathcal{C} \end{aligned} \tag{2}$$



All points on the line that passes through x_1, x_2

2. A **2-D affine set**. Suppose $x_1, x_2, x_3 \in \mathcal{C}$, then

$$\begin{aligned} \theta_1 x_1 + \theta_2 x_2 + (1 - \theta_1 - \theta_2) x_3 &\in \mathcal{C}, \\ \implies x_3 + \theta_1 (x_1 - x_3) + \theta_2 (x_2 - x_3) &\in \mathcal{C} \end{aligned} \tag{3}$$



All points on a plane that passes through x_1, x_2, x_3 .

Assume that an affine set $\mathcal{C}$ is a subset of $\mathbb{R}^3$. If there is another point $x_4 \in \mathcal{C}$ that does not belong to the plane, then $\mathcal{C} = \mathbb{R}^3$. Thus, the 2D affine set is the largest affine set in $\mathbb{R}^3$ that does not span the whole space. In $\mathbb{R}^n$, this will be called a hyperplane.

3. The set $\mathcal{C} = \{x | Ax = b\}$ is an affine set for any $x_1, \dots, x_k \in \mathcal{C}$

$$\begin{aligned} &\implies Ax_1 = b, Ax_2 = b, \dots, Ax_k = b \\ \implies A(\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k) &= (\theta_1 + \theta_2 + \dots + \theta_k)b = b \quad (4) \\ &\quad \forall \theta_1 + \theta_2 + \dots + \theta_k = 1 \end{aligned}$$

In $\mathbb{R}^3$, the line will correspond to 2 equations, and the plane will correspond to 1 equation. In $\mathbb{R}^n$, 1 equation (i.e., $Ax = b$) will be a hyper-plane.

Any affine set can be described by some linear equations (see later).